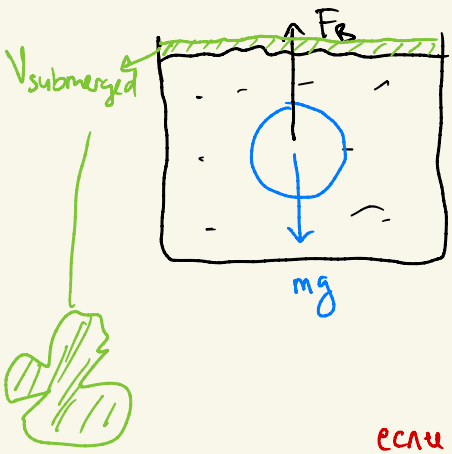
# Chapter 14: Fluids

- Поверхностное натяжение * (Resnick doesn't cover this)

- **Buoyant force**: Закон Архимеда




$F_B = ?$ $\rho_{liquid} V_{immersed} g$

$\Sigma F_{net} = ma$

$-mg + F_B = ma$

$$-\rho_{obj} g V_{im} + \rho_w g V_{im} = \rho V_{obj} a$$

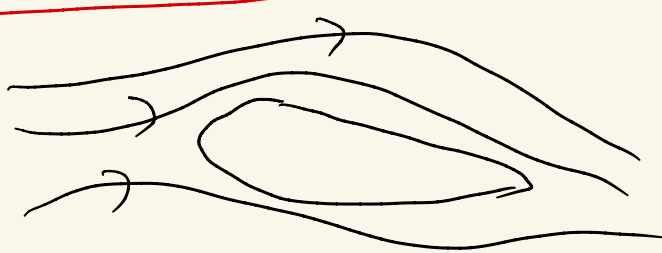
если полая структура $-mg + \rho_w g V_{im} = ma$

$m \neq \rho_{obj} V_{im}$

- Bernoulli Principle

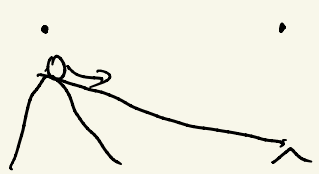
$$P_1 + \frac{1}{2}\rho v^2 + \rho g h = \text{const}$$

IPhO 2012 Theory

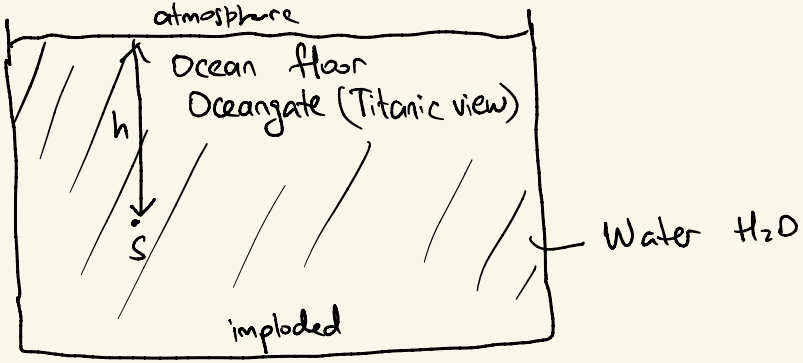


Pressure $P = \frac{F}{Area}$ High P low P

Drag force: $\frac{1}{2} C_A \rho v^2_{air} \propto kv^2$ (k = const)



In liquids



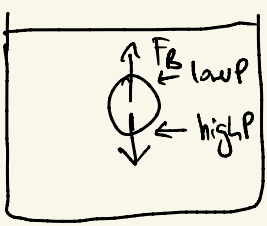

$$P(s) = P_{atm} + \underbrace{\rho_w g h}_{\text{why?}}$$

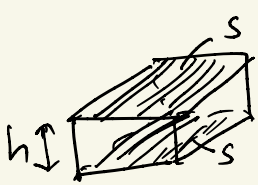
$$\text{Pressure} = \frac{\Delta F}{\Delta A} = P \quad \{\text{Pascals}\}$$

$[\text{atm}]$

$[\text{Torr}]$

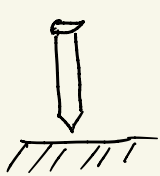
Q: Where does the Buoyant force originate?




$$mg = \rho g V = \rho g S h$$

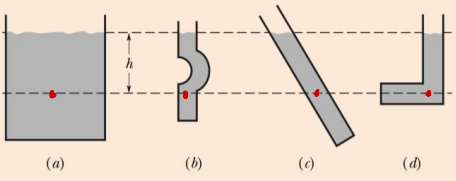
$$P = \frac{F}{A} = \frac{\rho g S h}{S} = \underbrace{\rho g h}$$



$P_{\text{blood}} = \frac{F}{S}$ ozemb Mongolbkaq

**CHECKPOINT 1**

The figure shows four containers of olive oil. Rank them according to the pressure at depth $h$, greatest first.




All SAME!

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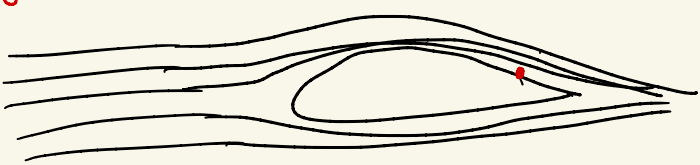
$$P_{a_1} + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 = \text{const.}$$

Во многих случаях $\updownarrow h$ same $\rho g h_1 \simeq \rho g h_2$

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2 = \text{const.}$$

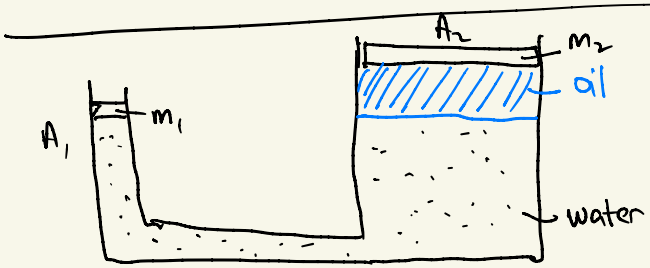
$P_2 \downarrow$ $\frac{1}{2}\rho v_2^2 \uparrow$

Observe: When $v$ = high, $P \downarrow$



check IPhO 2012 th

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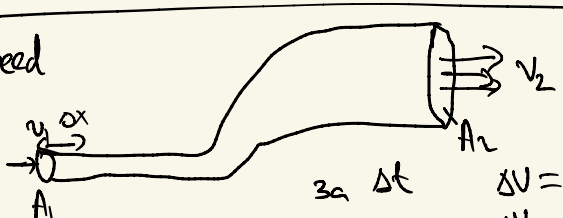

$$P_1 = P_2$$

$$\frac{m_1 g}{A_1} = \frac{m_2 g}{A_2} + \frac{\rho_{oil} V_{oil} g}{A_2}$$

---

① All flowing at same speed

② Incompressible : → Volume conserved




$A_1 v_1 = A_2 v_2$

$\Delta V = (\text{Area})\Delta x$

$\frac{\Delta V}{\Delta t} = A_1 \frac{\Delta x}{\Delta t} = A v$

## Problem T1. Focus on sketches *(13 points)*

### Part A. Ballistics *(4.5 points)*

A ball, thrown with an initial speed $v_0$, moves in a homogeneous gravitational field in the $x$-$z$ plane, where the $x$-axis is horizontal, and the $z$-axis is vertical and antiparallel to the free fall acceleration $g$. Neglect the effect of air drag.

**i. (0.8 pts)** By adjusting the launching angle for a ball thrown with a fixed initial speed $v_0$ from the origin, targets can be hit within the region given by

$$z \le z_0 - kx^2.$$

You can use this fact without proving it. Find the constants $z_0$ and $k$.

**ii. (1.2 pts)** The launching point can now be freely selected on the ground level $z = 0$, and the launching angle can be adjusted as needed. The aim is to hit the topmost point of a spherical building of radius $R$ (see fig.) with the minimal initial speed $v_0$. Bouncing off the roof prior to hitting the target is not allowed. Sketch qualitatively the shape of the optimal trajectory of the ball (use the designated box on the answer sheet). Note that the marks are given only for the sketch.




**iii. (2.5 pts)** What is the minimal launching speed $v_{\min}$ needed to hit the topmost point of a spherical building of radius $R$?



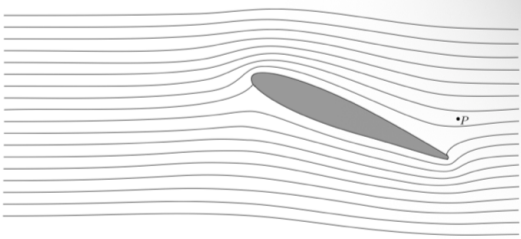
La Geode, Parc de la Villette, Paris. Photo: katchooo/flickr.com

### Part B. Air flow around a wing *(4 points)*

For this part of the problem, the following information may be useful. For a flow of liquid or gas in a tube along a streamline, $p + \rho g h + \frac{1}{2}\rho v^2 = \text{const.}$, assuming that the velocity $v$ is much less than the speed of sound. Here $\rho$ is the density, $h$ is the height, $g$ is free fall acceleration and $p$ is hydrostatic pressure. Streamlines are defined as the trajectories of fluid particles (assuming that the flow pattern is stationary). Note that the term $\frac{1}{2}\rho v^2$ is called the dynamic pressure.

In the fig. shown below, a cross-section of an aircraft wing is depicted together with streamlines of the air flow around the wing, as seen in the wing's reference frame. Assume that (a) the air flow is purely two-dimensional (i.e. that the velocity vectors of air lie in the plane of the figure); (b) the streamline pattern is independent of the aircraft speed; (c) there is no wind; (d) the dynamic pressure is much smaller than the atmospheric pressure, $p_0 = 1.0 \times 10^5\,\text{Pa}$.

You can use a ruler to take **measurements from the fig. on the answer sheet**.




**i. (0.8 pts)** If the aircraft's ground speed is $v_0 = 100\,\text{m/s}$, what is the speed of the air, $v_P$, at the point $P$ (marked in the fig.) with respect to the ground?

**ii. (1.2 pts)** In the case of high relative humidity, as the ground speed of the aircraft increases over a critical value $v_{\text{crit}}$, a stream of water droplets is created behind the wing. The droplets emerge at a certain point $Q$. Mark the point $Q$ in the fig. on the answer sheet. Explain qualitatively (using formulae and as little text as possible) how you determined the position of $Q$.

**iii. (2.0 pts)** Estimate the critical speed $v_{\text{crit}}$ using the following data: relative humidity of the air is $r = 90\%$, specific heat capacity of air at constant pressure $c_p = 1.00 \times 10^3\,\text{J/kg}\cdot\text{K}$, pressure of saturated water vapour: $p_{sa} = 2.31\,\text{kPa}$ at the temperature of the unperturbed air $T_a = 293\,\text{K}$ and $p_{sb} = 2.46\,\text{kPa}$ at $T_b = 294\,\text{K}$. Depending on your approximations, you may also need the specific heat capacity of air at constant volume $c_V = 0.717 \times 10^3\,\text{J/kg}\cdot\text{K}$. Note that the relative humidity is defined as the ratio of the vapour pressure to the saturated vapour pressure at the given temperature. Saturated vapour pressure is defined as the vapour pressure by which vapour is in equilibrium with the liquid.

# HW Ch 14

Problems: 24 (go through solution, it's a bit tricky)

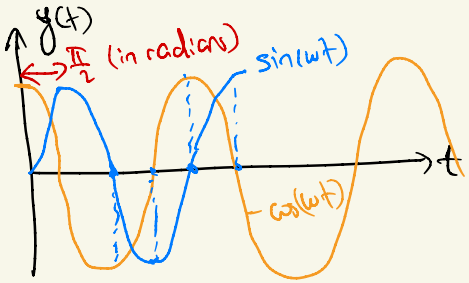
27, 28, 29, 32, 33, 41, 48,

58, 62, 64, 65-66, 67, 71, 83

Deadline: Thu 7th of August 12:00 pm noon

# Chapter 15: Oscillations

- cutty cosganuna : functions of $\cos(\omega t)$ and $\sin(\omega t)$



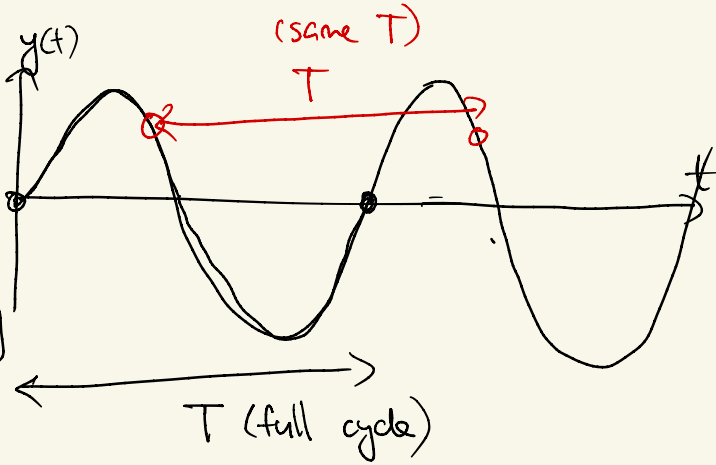

They're same

$$\cos(\omega t) = \sin\left(\omega t + \frac{\pi}{2}\right)$$

just an additional phase lag

$\pi = 3.14\ldots$ rads

$\pi = 180^\circ$ (degrees)




Period : $T$ (seconds)

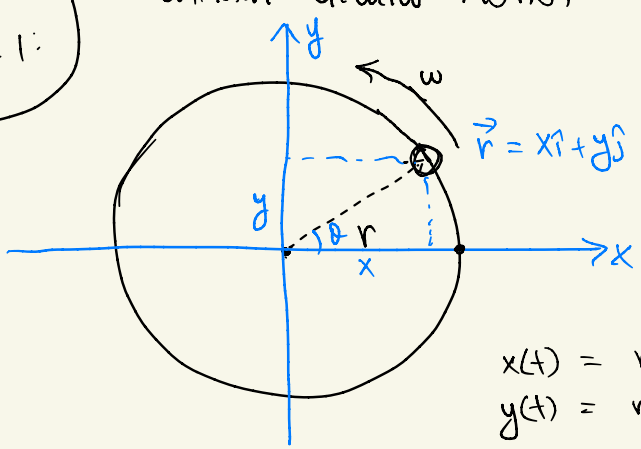
Frequency: $f = \frac{1}{T}$ (Hertz)

$[1\ \text{Hz} = 1\text{s}^{-1}]$

Ex 1:

Case 1:

Uniform circular Motion




$$|\vec{r}| = r = \sqrt{x^2+y^2}$$

$x = r\cos\theta$

$y = r\sin\theta$

Так как это движение

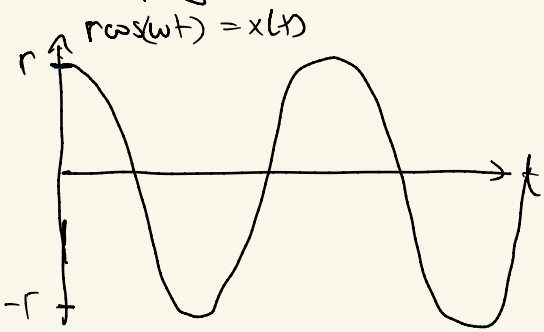
$\theta = \omega t$

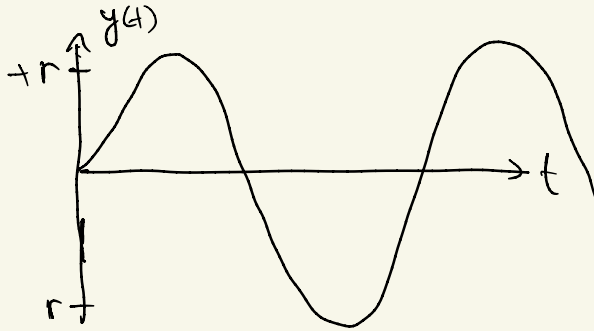
$x(t) = r\cos(\omega t)$, $x(t=0) = r\cos 0 = r$

$y(t) = r\sin(\omega t)$, $y(t=0) = r\sin 0 = 0$.

Если напишем уравн. $x(t)$ и $y(t)$

Нарисуем



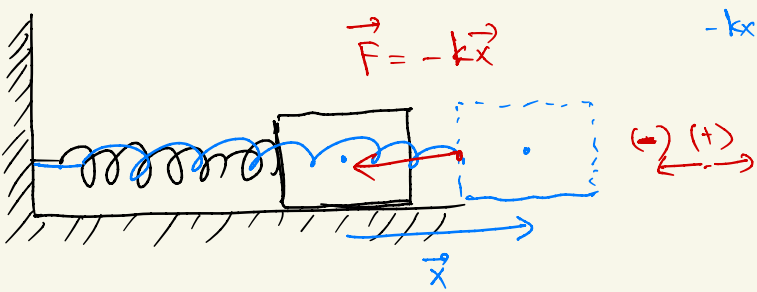




$$-1 \leq \cos, \sin \leq +1$$

$x(t)$, $y(t)$ по сути гармонические колебания (в фазах)

---

Ex2:




$$F = ma$$

$$-kx = m\frac{d^2x}{dt} = m\ddot{x}$$

$$\ddot{x} + \frac{k}{m}x = 0$$

$$\ddot{x} + \omega^2 x = 0$$

уравнение SHO (simple harmonic osc)

2nd Order Homogeneous Ordinary Diff. Eq.

$\omega$: oscillation frequency

$$x(t) = A\cos(\omega t) + B\sin(\omega t)$$

The most general solution to $\ddot{x} + \omega^2 x = 0$

$$\frac{dx}{dt} = -\omega A\sin(\omega t) + \omega B\cos(\omega t)$$

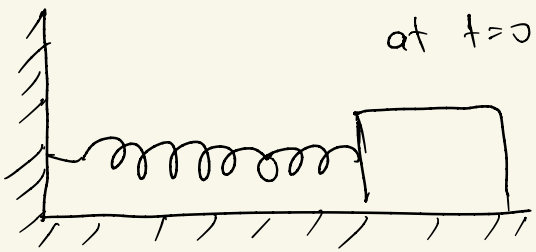
$$\frac{d^2x}{dt^2} = -\omega^2 A\cos(\omega t) - \omega^2 B\sin(\omega t)$$

$$\ddot{x} + \omega^2 x = -\omega^2(A\cos\omega t + B\sin\omega t) + \omega^2 x(t) = 0$$ ✓

SHO Eqn: $\frac{d^2x}{dt^2} + \omega^2 x = 0$

General soln: $\boxed{x(t) = A\cos(\omega t) + B\sin(\omega t)}$

Ex: $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \rightarrow \omega = \sqrt{\frac{k}{m}}$ ↓ reduces to



at $t=0$, $x_0=0$

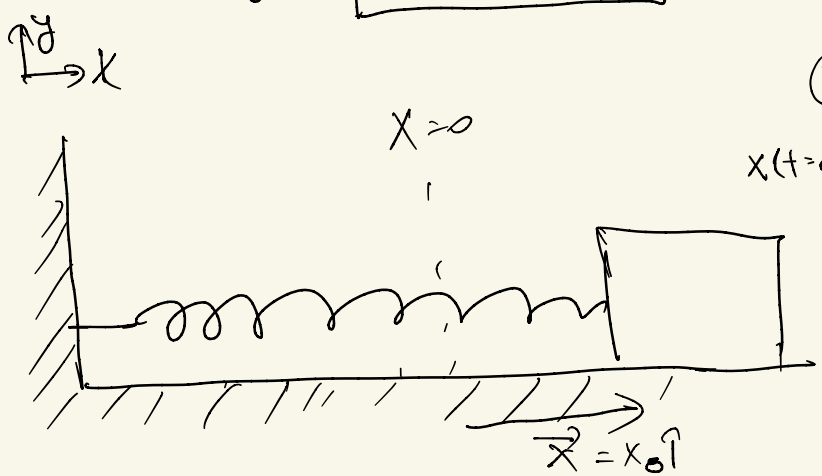
$\boxed{x(t) = X_0\cos(\omega t+\phi)}$ for mass spring system

- A ve B hangisi iş initial conditions

(1) Ecm $x(t=0) = 0$

$x(t=0) = A\cos(0) + B\sin(0) = A \rightarrow 0.$

Torga $\boxed{x(t) = B\sin(\omega t)}$

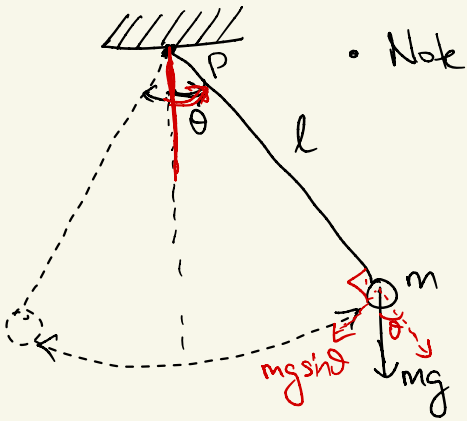



(2) Ecm $x(t_0) = +x_0$

$x(t=0) = A\cos 0 + B\sin 0 = x_0$

$\boxed{A = x_0}$

SHO Ex3:




• Note θ is small

θ - 15°

this simplifies our task (eqn) a lot!

$\vec{\tau} = I\alpha$

$I = ml^2$

$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$

$\vec{\tau} = \vec{r} \times \vec{F}$

$= |\vec{r}||\vec{F}|\sin\theta$

$= r_\perp F$

$-mg\sin\theta \cdot l = I\frac{d^2\theta}{dt^2}$

small angle approximation

$\sin\theta \simeq \theta$ (in radians)

$-mgl\theta = I\ddot{\theta} = ml^2\ddot{\theta}$

$\ddot{\theta} + \frac{mgl}{ml^2}\theta = 0$

$\ddot{\theta} + \frac{g}{l}\theta = 0$

Remember?

$\ddot{x} + \omega^2 x = 0$

$\omega = \sqrt{\frac{g}{l}}$

SHO eqn

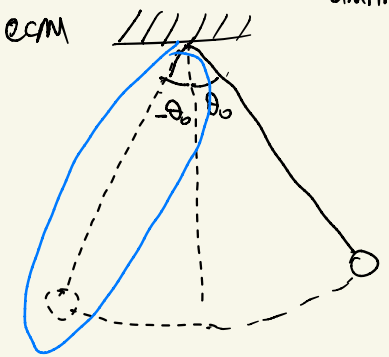
$\theta(t) = \theta_0 \cos(\omega t + \phi)$

★ General

$\theta(t) = \theta_0 \cos(\omega t + \phi)$

$\theta_0$ ↑ амплитуда

$\phi$ ↑ найдём с начальных условий




Ex! начальные условия: $\theta(t=0) = -\theta_0$

What should be $\phi$ = ?

$\theta(t=0) = \theta_0 \cos(\omega \cdot 0 + \phi)$

$= \theta_0 \cos\phi$

$= -\theta_0 \longrightarrow \cos\phi = -1 \rightarrow \boxed{\phi = \pi}$

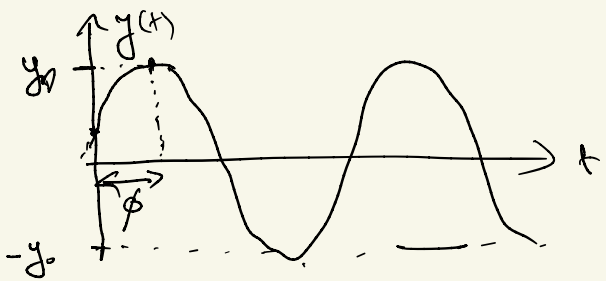
Ex2: $\omega = \frac{2\pi}{T}$, at $t = \frac{T}{2}$, $\theta = -\theta_0$, what $\phi$ = ? satisfies this

$\theta(t=\frac{T}{2}) = \theta_0 \cos(\frac{2\pi}{T}\cdot\frac{T}{2} + \phi) = -\theta_0$, $\cos(\pi+\phi) = -1$, $\boxed{\phi = 0}$

---

In general, in harmonic oscillation $y(t) = y_0 \cos(\omega t + \phi)$

$y_0$ ↑ amplitude, $\phi$ ↑ phase




if $y(t=0) = y_0$

Тогда $\cos(\omega t + \phi) = 1$

$\cos(\phi) = \cos(0)$

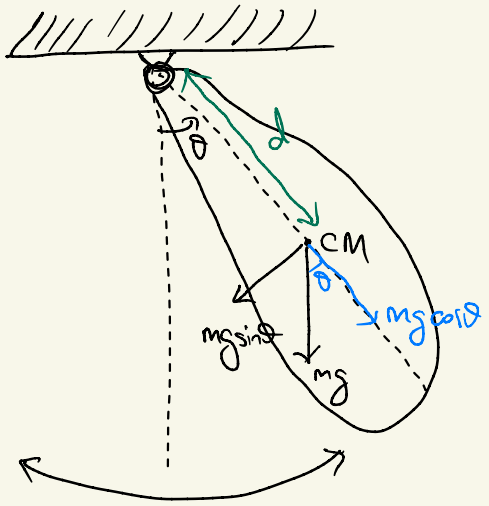
$\phi = 0$

$\omega = \sqrt{\frac{g}{l}}$, $\omega = 2\pi f = \frac{2\pi}{T} = \sqrt{\frac{g}{l}}$

$$\boxed{T = 2\pi\sqrt{\frac{l}{g}}}$$

$$\boxed{f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}}$$

SHO EX4: Physical Pendulum




$$\tau = I\alpha$$

$$-mg\sin\theta d = I\ddot{\theta}$$

$$-mg\theta d = I\ddot{\theta}$$

$$\ddot{\theta} + \frac{mdg}{I}\theta = 0$$

$$\omega = \sqrt{\frac{mgd}{I}}$$

$\omega$ = angular frequency

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$f = \frac{1}{T}$$

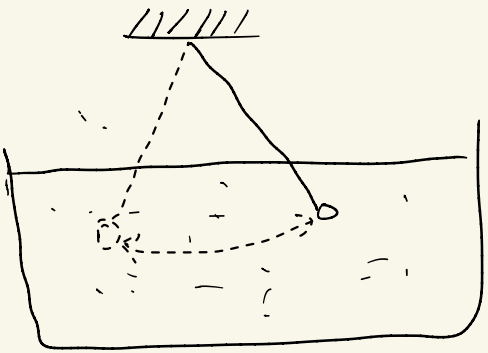
$$\omega = \frac{2\pi}{T} = \sqrt{\frac{mgd}{I}}$$

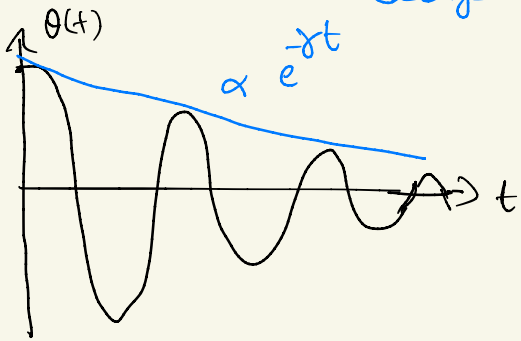
$$f = \frac{1}{2\pi}\sqrt{\frac{mgd}{I}}$$

$$T = 2\pi\sqrt{\frac{I}{mgd}}$$

# Damped (Driven) oscillations

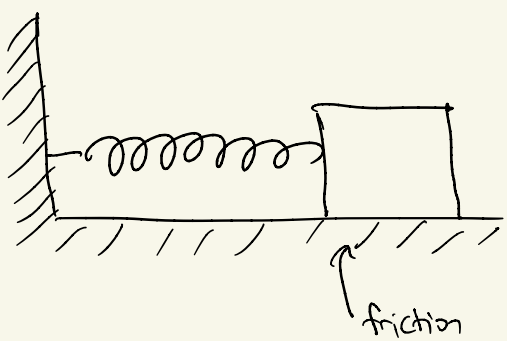
ex1: in Viscous medium





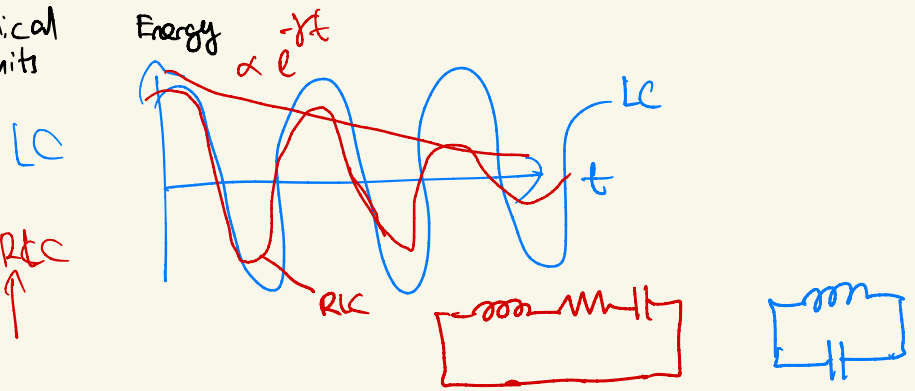

exponentially decays

ex2:




or air drag $\propto v^2$

Electrical Circuits




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THINGS TO LEARN (later, Сибухин - Савченко)
in Damped Oscillations

Remember

1) Exponential decay
2) Quality factor Q: Amplitude $A \to \frac{A}{e}$ time

# HW Ch 15

Questions: 1, 5, 7 (Answers are at the Appendix of the Book)

Problems: 3, 9, 11, 22, 24, 25, 26, 33, 34

$a = \frac{d^2x}{dt^2} = -\omega^2 x_0 \cos(\omega t + \phi)$

$-1 \leq \cos \leq 1$ for max. $\cos = 1$

41, 48 - (just understand the solution after giving a try)

51, 52, 53, 54, 56, 100

Deadline: Thu 7th of August 12pm noon